Course Outline for EE-183

Part A

1. Course Code: EE-183

2. Course Title: Electronic Devices and Circuits

3. Course Type: Core Course

4.Level/ Term: Level: 1 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Annesha Das, Assistant Professor, Dept. of CSE, CUET and
Ashim Dey, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 4

9. Contact Hours: 4 lectures of 50 minutes class per week

10. Total Marks: 400

11. Rational of the Course:

This course will cover the design of different digital electronics circuits and different pulse generations circuits. This course is important to understand how linear and nonlinear wave shaping circuits; pulse generating circuit such as multivibrator; DAC-ADC; oscillator; amplifier works. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Introduction to semiconductors; Bipolar Junction Transistor (BJT): *Bipolar Junction Transistor (BJT): principle of operation, I-V characteristics; Transistor circuit configurations (CE, CB, CC), BJT biasing, load lines; BJTs at low frequencies; Hybrid model, h parameters, simplified hybrid model; Small-signal analysis of single and multi-stage amplifiers, frequency response of BJT amplifier.* ***Field Effect Transistors (FET):*** *principle of operation of JFET and MOSFET; Depletion and enhancement type NMOS and PMOS; biasing of FETs; Low and high frequency models of FETs, Switching circuits using FETs;* ***Introduction to CMOS; OPAMP:*** *linear applications of OPAMPs, gain, input and output impedances, active filters, frequency response and noise, Non-linear applications of OP AMP;* ***DAC-ADC:*** *A/D and D/A converters with applications; Sample and hold*

circuit; ***LED-LCD:*** *Introduction of LED and LCD with application; Oscillators; Silicon Controlled Rectifiers (SCR), TRIAC, DIAC and UJT: characteristics and applications; Introduction to IC fabrication processes. Stabilizer and UPS.*

Wave shaping circuits: *Linear wave shaping; Non-linear wave shaping; Diode wave shaping technique s, clipping and clamping circuits (Diode applications); regulated power supply using zener diode; comparator circuits;* ***Multivibrators:*** *monostable, bistable and astable multivibratiors; Schmitt trigger; blocking oscillators and time-base circuit; Timing circuits, Simple voltage sweeps, linear current sweeps.*

12. Course Objectives:

(a) Introduce with different digital electronics circuits and different pulse generations
(b) Design and analyze different electronic devices & circuits and find their advantages disadvantages.
(c) Analyze working operations of linear & nonlinear wave shaping and pulse generating circuits.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Describe operations, working principles and implement different electronic devices.	C1,C2,C6,P1
CLO2	Develop skills and knowledge in analysis and design of electronic circuits.	C5
CLO3	Design and analyze different linear and nonlinear wave shaping circuit, and pulse generating circuits.	C5,P7
CLO4	Analyze different converter, oscillators and amplifiers and learn their applications	C4

b) Mapping of CLO with PLO

No.	(CLOs)	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO1 0	PLO 11	PLO 12
1	CLO1	X											
2	CLO2	X		X									
3	CLO3	X	X	X									
4	CLO4	X	X										

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	***Introduction to semiconductors:*** *Intrinsic and extrinsic semiconductor, conductivity of extrinsic semiconductor, n-type and ptype semiconductor, majority and minority carrier, P-N junction: formation and properties, current flow in a p-n junction, biasing of p-n junction*	• Lecture	Class Test Final Exam	CLO-1
Week - 02	***Bipolar Junction Transistor (BJT):*** *Bipolar Junction Transistor (BJT): principle of operation, BJT biasing, load lines; BJTs at low frequencies;*	• Lecture	Class Test Final Exam	CLO-1
Week - 03	*Transistor circuit configurations (CE, CB, CC), I-V characteristics*	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1 CLO-2

Week - 04	*Hybrid model, h parameters, simplified hybrid model; Small-signal analysis of single and multi-stage amplifiers, frequency response of BJT amplifier.*	• Lecture	Class Test Final Exam	CLO-2
Week - 05	***Field Effect Transistors (FET):*** *principle of operation of JFET and MOSFET; Depletion and enhancement type NMOS and PMOS; biasing of FETs; Low and high frequency models of FETs, Switching circuits using FETs,* ***Introduction to CMOS***	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week - 06	*Silicon Controlled Rectifiers (SCR), TRIAC, DIAC and UJT: characteristics and applications; Introduction to IC fabrication processes. Stabilizer and UPS*	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1 CLO-2
Week - 07	*Linear wave shaping (High pass RC, Low pass RC)*	• Lecture	Class Test Final Exam Assignments	CLO-3
Week - 08	*Non-linear wave shaping; Diode wave shaping technique s, clipping and clamping circuits (Diode applications); regulated power supply using zener diode; comparator circuits*	• Lecture	Class Test Final Exam	CLO-3
Week - 09	***Multivibrators:*** *monostable, bistable and astable multivibratiors; Schmitt trigger; blocking oscillators*	• Lecture	Class Test Final Exam Assignments	CLO-3
Week - 10	***Time-base circuit****; Timing circuits, Simple voltage sweeps, linear current sweeps;* ***LED-***	• Lecture • Assignments	Class Test Final Exam	CLO-3

	LCD: *Introduction of LED and LCD with application*		Assignments	
Week - 11	***Operational amplifiers****: OP-Amp basics, inverting and non-inverting amplifiers, summing point, Unity follower, adder, subtractor, comparator, integrator, differentiator, active filters*	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week - 12	***DAC-ADC:*** *A/D and D/A converters with applications; Sample and hold circuit*	• Lecture	Class Test Final Exam	CLO-4
Week - 13	Review Class			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests/Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 30%

b) Summative: 70%

c) Make-up Procedures:

- Feedback on continuous assessment is given to the students immediately after the test.
- The minimum number of class-test/assignment are (n+1) with best n will be counted (here, n is number of credit). Based on the students' feedback additional class-test/assignment may be taken by the course teacher

Part D

16. Learning Materials

1) Recommended Readings

- A Textbook of Electrical Technology by B.L. Theraja (Volume IV)
- Handbook of Electronics Writer: Kumar Gupta
- Pulse, Digital and Switching Waveforms by Jacob Millman and Herbert Taub

2) Supplementary Readings

- Electronic Devices by Thomas L. Floyd
- Principles of Electronics by V.K. Mehta and Rohit Mehta
- Digital Systems: Principles and Applications by Ronald J. Tocci and Neal S. Widmer